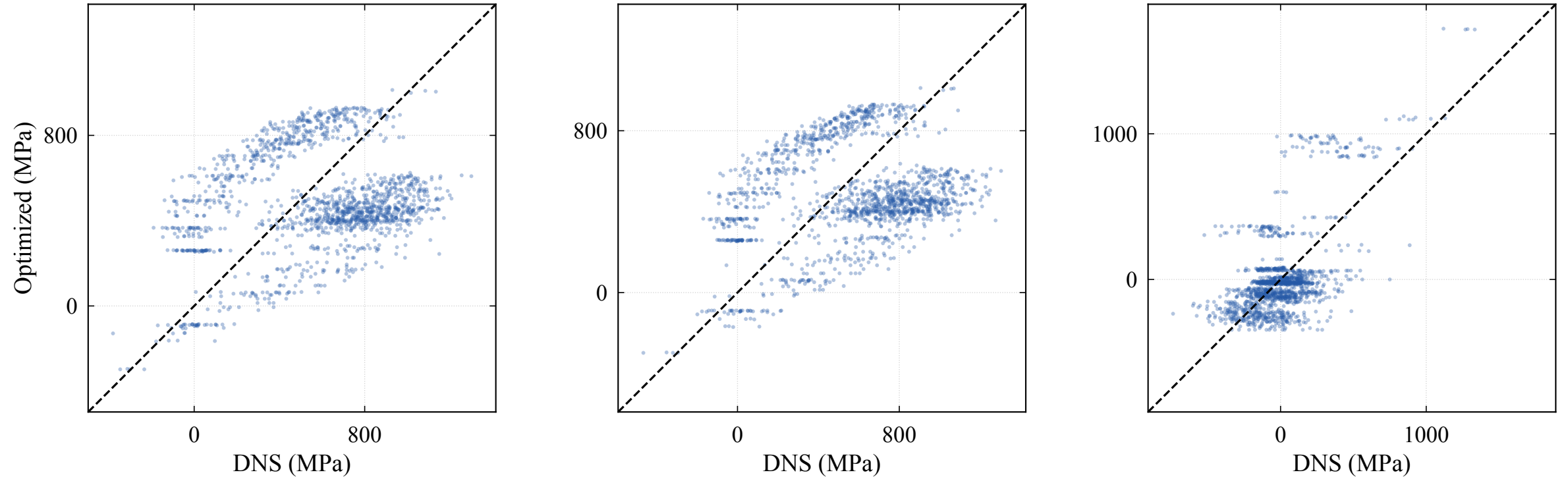
DNS vs optimized: paired-field agreement

eL2 = 55.3% | R² = -0.27
RMSE = 368 MPa

eL2 = 55.7% | R² = -0.26
RMSE = 373 MPa

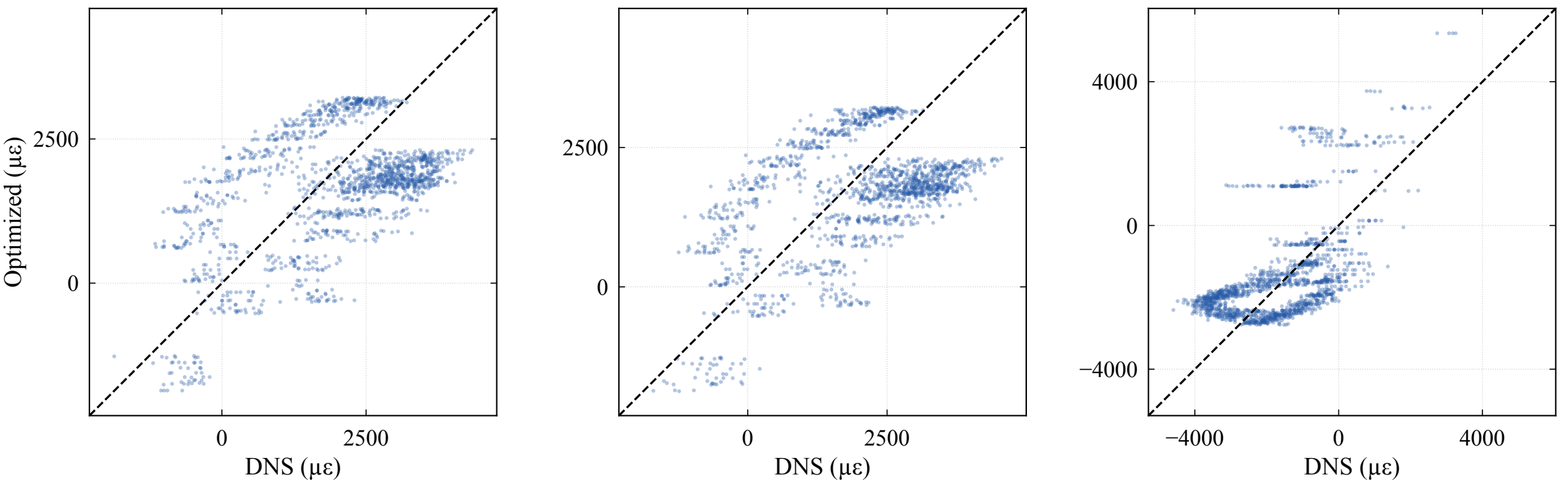
eL2 = 108.8% | R² = -0.18
RMSE = 238 MPa



(a) Stress xx
eL2 = 52.1% | R² = 0.11
RMSE = 1.22e+03 με

(b) Stress yy
eL2 = 52.8% | R² = 0.10
RMSE = 1.24e+03 με

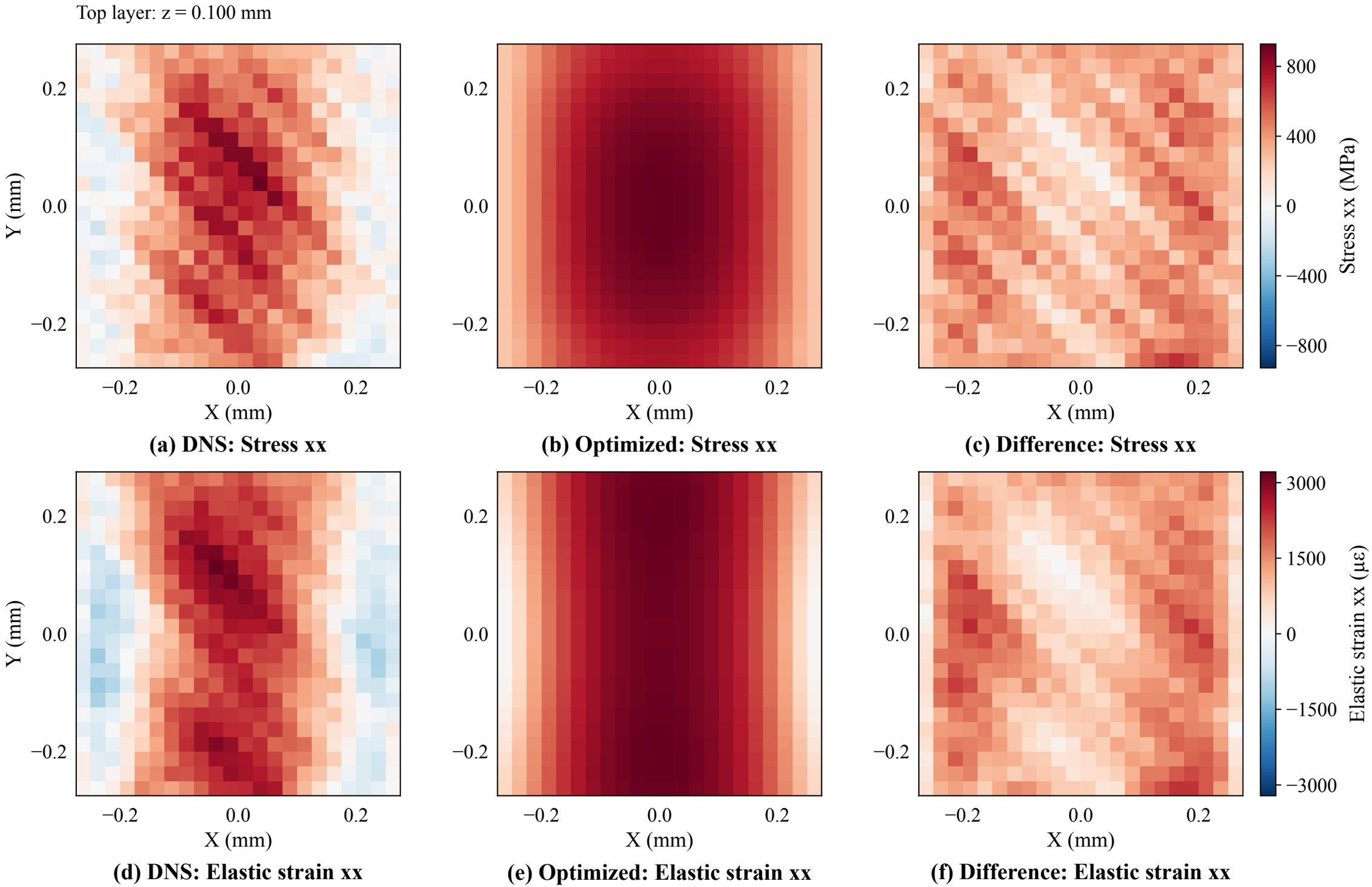
(c) Stress zz
eL2 = 61.1% | R² = 0.05
RMSE = 1.31e+03 με



(d) Elastic strain xx

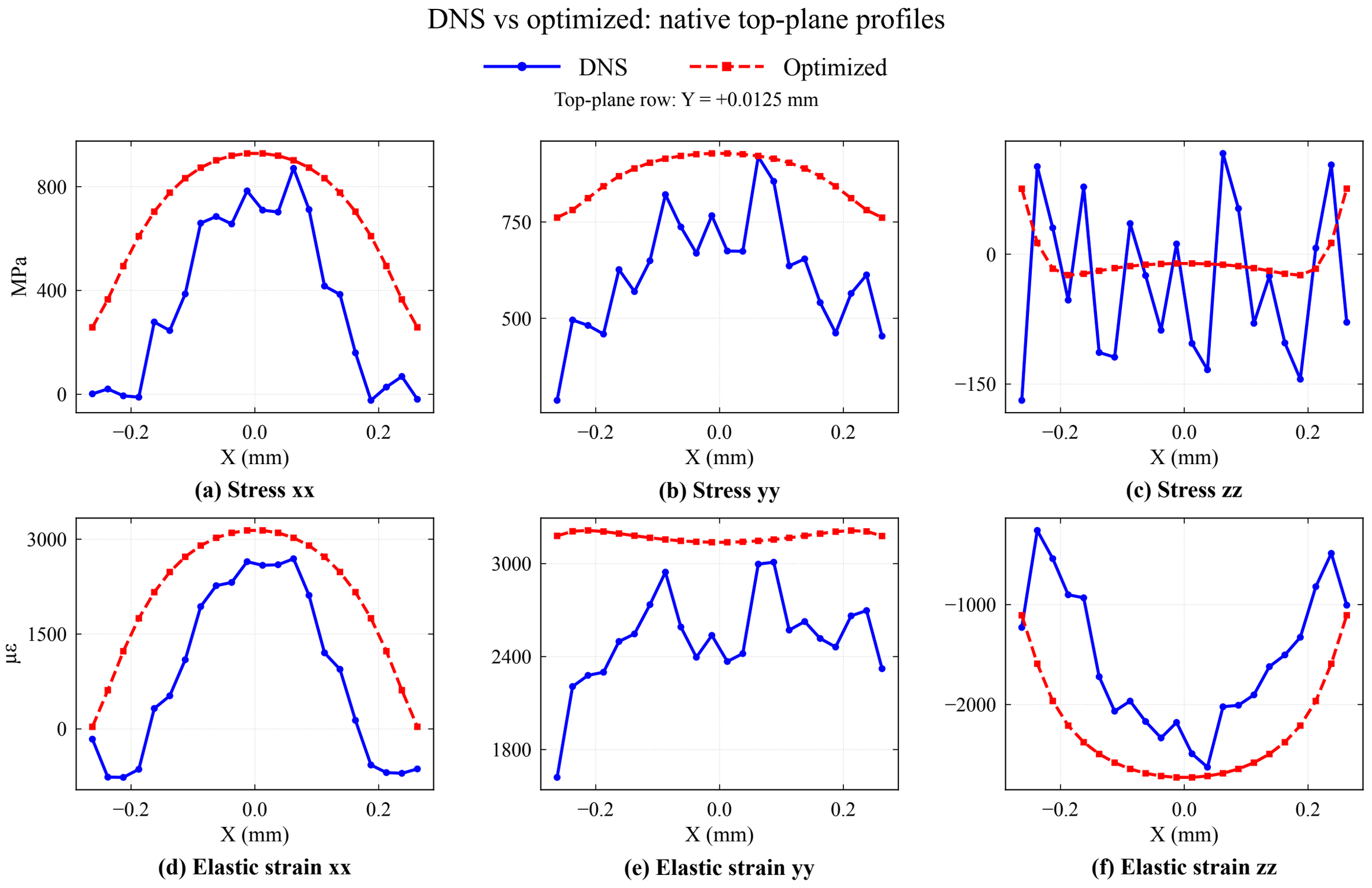
(e) Elastic strain yy

(f) Elastic strain zz



(a) DNS: Stress xx **(b) Optimized: Stress xx** **(c) Difference: Stress xx**

(d) DNS: Elastic strain xx **(e) Optimized: Elastic strain xx** **(f) Difference: Elastic strain xx**


DNS vs optimized: native top-plane profiles
DNS
Optimized
Top-plane row: Y = +0.0125 mm
MPa
800
400
0
−0.2
0.0
0.2
X (mm)
(a) Stress xx
750
500
X (mm)
(b) Stress yy
0
−150
X (mm)
(c) Stress zz
με
3000
1500
0
X (mm)
(d) Elastic strain xx
3000
2400
1800
X (mm)
(e) Elastic strain yy
−1000
−2000
X (mm)
(f) Elastic strain zz